3. Solve for the unknown:

a) $-5 + 5(n - 7) = -40 + 5n$

c) $4x(x + 10) = 0$

d) $5 - \frac{2(m+4)}{m} = \frac{7}{m}$

e) $\frac{x+2}{x} = \frac{x-1}{x} - \frac{4x+2}{x^2 - 3x}$

f) $g = \frac{x - c}{x}$ $for\ x$

g) $(x - 3)(-ax + 2c) = 0$ $for\ x$

Question 4

a) Given: $4y - 2x = 8$ and $x + 2y = 8$

i) Will (0; 2) be a solution for both equations? Give a reason for your answ

b) Solve for the unknown: $\frac{1}{r+3} = \frac{r+4}{r-2} + \frac{6}{r-2}$

c) Given $\frac{x}{3x^2+2x-1}$,

i) For which value(s) of x will the fraction = 0?

ii) Which value(s) of x will make the fraction undefined?

30) $-5 + 5n - 35 = -40 + 5n$

$5n - 40 = 5n - 40$

$0 = 0$

$\therefore\ n \in \mathbb{R}$.

Solve for the unknown:

a) $-5 + 5(n - 7) = -40 + 5n$

c) $4x(x + 10) = 0$

c) $4x = 0$ or $x + 10 = 0$

$x = 0$ or $x = -10$

d) $5 - \frac{2(m+4)}{m} = \frac{7}{m}$

d) LCD $= m$ $\quad m \neq 0$

$5m - 2(m + 4) = 7$

$5m - 2m - 8 = 7$

$3m = 15$

$m = 5$

f) $g = \frac{x - c}{x}$ for x

f) $gx = x - c$

$gx - x = c$

$x(g - 1) = c$

$x = \frac{c}{g - 1}$

e) $\frac{x+2}{x} = \frac{x-1}{x} - \frac{4x+2}{x^2 - 3x}$

g) $(x - 3)(-ax + 2c) = 0$ for x

e) LCD $= x(x - 3)$ $\quad x \neq 0;\ x \neq 3$

$(x + 2)(x - 3) = (x - 1)(x - 3) - (4x + 2)$

$x^2 - x - 6 = x^2 - 4x + 3 - 4x - 2$

$7x = 7$

$x = 1$

g) $x = 3$ or $-ax + 2c = 0$

$-ax = -2c$

$x = \frac{2c}{a}$ $\quad a \neq 0$

Question 4

4ai)

a) Given: $4y - 2x = 8$ and $x + 2y = 8$

i) Will (0; 2) be a solution for both equations? Give a reason for your answ

$4y - 2x = 8$

Sub (0; 2)

LHS $= 4(2) - 2(0)$

$= 8$

$=$ RHS

$x + 2y = 8$

LHS $= 0 + 2(2)$

$= 4$

$\neq 8$ ∴ not a soln. of both equations.

b) $\frac{1}{r+3} = \frac{r+4}{r-2} + \frac{6}{r-2}$

b) Solve for the unknown: $\frac{1}{r+3} = \frac{r+4}{r-2} + \frac{6}{r-2}$

$r \neq -3$; $r \neq 2$

$r - 2 = (r+3)(r+10)$

$r - 2 = r^2 + 13r + 30$

$r^2 + 12r + 32 = 0$

$(r+8)(r+4) = 0$

$r = -8$ or -4

c) Given $\frac{x}{3x^2+2x-1}$,

i) For which value(s) of x will the fraction = 0?

ii) Which value(s) of x will make the fraction undefined?

ci) $x = 0$

ii) $3x - 1 = 0$

$x = \frac{1}{3}$

or $x = -1$

$\frac{x}{(3x^2 + 2x - 1)}$

$= \frac{x}{(3x-1)(x+1)}$